D04 · Supplier Selection & Qualification

# 05 · Onboarding & first-article approval

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

## 1 WHAT IT IS

Onboarding & first-article approval is the bridge from contract signature to first commercial PO — supplier enrollment in buyer systems, banking and tax setup, ERP/EDI integration, PPAP or FAI execution, and the validation that the qualified supplier can actually produce what the contract promises in the buyer's own conditions. It is the operational gate that turns a paper relationship into a producing one.

## 2 WHY IT MATTERS

The gap between "qualified" and "producing" is where most supplier relationships start badly. **A supplier that passed qualification on the audit floor in optimal conditions is not the same as a supplier producing at volume against the buyer's specs, on the buyer's schedule, with the buyer's quality acceptance criteria**. The standard reference work documents this clearly **(AIAG PPAP 4th Edition, 2006 — automotive industry standard, widely adopted across aerospace and medical)**: **first-article approval is the structured verification that closes the gap between intent and execution**, and the operations that skip it discover the gap on the first production batch — usually when it's too late to do anything but pay the cost.

## 3 HOW IT IS DONE

1. **Build the onboarding checklist with explicit owners — not a generic process flow.** Banking setup (treasury), tax forms (finance), ERP master data (IT), EDI configuration (supply chain IT), quality plan (quality), logistics setup (logistics). Each step has a named owner with a target completion date. Without that explicit assignment, onboarding stalls in handoffs.
2. **Run PPAP / FAI on the first production tooling and the first commercial volume.** Many operations PPAP the prototype and skip PPAP on the production-equivalent run; the discovery is that prototype conditions and production conditions differ in ways that affect quality. PPAP must verify the actual production process at production rate with production tooling.
3. **Validate ERP/EDI integration end-to-end with test transactions before go-live.** A PO that the supplier cannot acknowledge electronically, an ASN that fails the buyer's system validation, an invoice that does not match the three-way match logic — all common, all preventable by structured integration testing before the first commercial transaction.
4. **Build the ramp-up curve explicitly — first PO is not target volume.** A new supplier needs production volume to find their issues at scale; jumping from PPAP to full target volume is the common cause of early quality excursions. Define an explicit ramp curve over 4–12 weeks depending on category complexity.
5. **Establish the early performance review cadence — weekly for the first quarter.** Quality data, OTIF, defect categories, response time — reviewed weekly with the supplier's account team during the ramp period. The cadence catches drift before it becomes a pattern; without it, problems emerge silently and surface as crises.
6. **Sign off onboarding formally — and only then move the supplier to standard governance.** Onboarding is closed when the named cross-functional team signs off that the supplier is producing at quality, at rate, with stable systems integration. Until then, the supplier stays in heightened oversight; closing onboarding prematurely is the structural error that creates the supplier nobody manages until they fail.

**WORKED EXAMPLE — ILLUSTRATIVE**

An automotive Tier 1 onboards a newly qualified supplier for a precision stamping component — **annual volume 4.8M parts, ramp from 0 to full rate over 8 weeks**. The onboarding plan: **(1) cross-functional kickoff** with 9 owners across IT, treasury, tax, quality, supply chain, logistics, manufacturing, engineering, and procurement — each with named deliverable and target date; **(2) banking, tax, ERP, EDI complete in 18 days**; **(3) PPAP submission** includes Level 3 documentation, 300-part capability study (Cpk > 1.67 on all critical characteristics), MSA on gauges, control plan; **(4) ramp schedule**: week 1-2 at 25% rate, week 3-4 at 50%, week 5-6 at 75%, week 7-8 at 100%; **(5) weekly performance review** with the supplier's plant manager and account team during the 8-week ramp + 12-week stabilization period; **(6) formal onboarding closure** at week 20 after demonstrating sustained quality (PPM < 200, FPY > 99.4%) at full rate. Two issues caught and resolved during ramp: a tooling break-in pattern producing dimensional drift in weeks 3-4 (resolved with adjusted maintenance cadence), and an ASN format incompatibility causing 8% of receipts to fail three-way match (resolved with EDI mapping fix). **Both would have surfaced as production crises at full rate without the structured ramp**; instead, they were caught and corrected during the protected period.

**IN PRACTICE**

### Build the onboarding checklist with named owners

Banking setup, tax forms, ERP master data, EDI integration, quality docs, first PO conditions — each step has a different owner, and a step without a named owner sits idle. The discipline of a checklist with a named owner per row converts onboarding from a string of handoffs into a tracked, accountable workflow.

### Run PPAP/FAI at production conditions, not pilot

A first article produced under hand-finished pilot conditions does not validate the production process; it validates the engineering team's ability to build a single good part. PPAP and FAI exist to verify the supplier's actual production setup — tooling, operators, sample size, statistical evidence — and the gate must close on that production evidence, not on a demonstrator.

### Validate ERP/EDI integration before go-live

Master data mismatches, EDI message format errors, tax code differences, unit-of-measure conversions — all of them surface on the first transaction if not tested. A structured integration test with a closed set of test transactions, run before the first commercial PO, is the only way to keep the relationship's first month from being spent debugging master data.

### Build the ramp curve explicitly, with review cadence

A supplier going from zero to full volume in week one is a supplier guaranteed to fail somewhere along the way. Define the ramp curve — week 1 at 10%, week 4 at 50%, week 12 at 100% — with weekly review meetings during ramp. The ramp curve is the protected period where issues are still cheap to fix; standard governance starts only after onboarding formally closes.

**WHAT YOU WOULD USE**

**ONBOARDING / SUPPLIER LIFECYCLE TOOLS**

**SAP Ariba SLP / Coupa SIM / GEP Smart** — Supplier lifecycle management modules with onboarding workflows, document repositories, and milestone tracking — the integrated approach when the procurement suite is already in place **(SAP — vendor; Coupa — vendor; GEP — vendor)**.

**QualityOne, ETQ Reliance, AssurX (PPAP modules)** — Specialized quality management platforms with structured PPAP/FAI workflows, capability studies, and approval cadences — strongest for automotive, aerospace, and medical device operations **(ETQ — vendor)**.

**EDI integration platforms (OpenText, Cleo, IBM Sterling)** — Structured B2B integration for the EDI/API setup with new suppliers — required when supplier electronic exchange is part of standard operations **(OpenText — vendor; Cleo — vendor)**.

**Structured onboarding checklist + named-owner tracker** — For operations not at scale — a structured checklist with owners, target dates, and weekly status review. Tooling matters less than the discipline of named ownership and target completion dates.

## 4 SO WHAT

**The signed contract is not an active supplier; the approved first article is.** Build the onboarding checklist with named owners, run PPAP/FAI at production conditions, validate ERP/EDI integration before go-live, build the ramp curve explicitly, establish weekly review cadence during ramp, and close onboarding formally before moving to standard governance. The plants that invest in structured onboarding catch the issues in the protected period; the plants that skip it discover the issues as production crises.